

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

Based on the incident, the browser attempts to send a request to DNS server through the UDP protocol to obtain the IP address of the domain "www.yummyrecipesforme.com". The ICMP echo reply returned as "udp port 53 unreachable". Port 53 in the error message is used for DNS service(Domain Name Server). Therefore this incident "port unreachable" is due to no service after the incoming DNS port.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

This event occurred at 1.24pm and 32.192571 seconds according to the tcpdump timestamp. Several customers and clients have first encountered this error message saying "destination port unreachable". When I visited the website "www.yummyrecipesforme.com" it took a while to load the webpage, and I got the message "destination port unreachable". After encountering this issue, I have used "tcpdump"/network protocol analyzer to see if I can resolve this issue. To take further action, our step is to reload the website "www.yummyrecipesforme.com" and apply tcpdump to see what errors are present. Overall the network security team has dealt by running tcpdump. After analyzing in detail, the tcpdump tells us that port 53, which is associated with the DNS server, where the UDP protocol takes the IP address to domain. Now we are investigating the root cause of this error, and understanding why the port number is wrong, even though the right IP address "UDP" is used to get to the DNS server.

Based on the incident, the IT department investigation has found that in order to load the webpage, the browser needs to send a query to a DNS server via the UDP protocol to retrieve the IP address for the website's domain name, and this is part of the DNS protocol. That later affected port 53 which is a port for DNS service, the word "unreachable" in the message indicates the UDP message requesting an IP

address for the domain “www.yummyrecipesforme.com”.

After extensive inspection on this incident, a likely cause of this incident is due to it not go through to the DNS server because no service was listening on the receiving DNS port.